Ishfaq Ahmad, Ph.D.

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Research Interests

- **Medical Decision Making & Screening Policy**
  Developing simulation-based natural history models to quantify the population-level benefits and harms of cancer screening, including early detection, overdiagnosis, and risk stratification, to inform national screening policy.
- **Mathematical Epidemiology & Disease Dynamics**
  Developing mechanistic models to study disease transmission dynamics and evaluate interventions, and targeted control strategies, to guide public health decision-making.
- **Integrated Modeling for Population Health Decision-Making**
  Applying complementary methodologies from infectious disease and cancer modeling to address complex population health challenges and generate evidence that directly informs screening policy and clinical guidelines.

Methods & Technical Skills

Modeling Frameworks	Microsimulation, Continuous-time Markov chains, Stochastic state-transition modeling, Compartmental / ODE modeling, System dynamics.
Analytical Techniques	Model calibration & validation, Parametric survival modeling, Scenario & sensitivity analysis, Factorial scenario design, Benefit harm evaluation.
Data Sources	SEER, SEER-Medicare, National Center for Health Statistics (NCHS) life tables, national surveillance data, clinical trial data (NLST, PLCO).
Programming & Tools	R (package development), Quarto, Python, MATLAB, Mathematica, High-performance / parallel computing (SLURM, cluster job arrays).
Scientific Communication	Overleaf/LaTeX, Scientific WorkPlace, PowerPoint, Microsoft Office.

Postdoctoral Experience

Mar 2024 – present

Postdoctoral Scholar, Knight Cancer Institute, Oregon Health & Science University, Portland, USA

Supervisor: Jane Lange (Staff Scientist) & Shelley Tworoger (Professor)

- Developed and validated **MCEDsim**, an R-based natural history microsimulation framework using continuous-time Markov chains to model disease progression and screen testing.
- Evaluated multi-cancer early detection strategies to quantify population-level trade-offs between screening benefits and harms (overdiagnosis, false positives) to inform policy.
- Developing a lung cancer natural history microsimulation model to quantify the contributions of LDCT screening uptake, adherence, and novel treatment access to racial/ethnic disparities in outcomes.

Mar 2023 – Feb 2024 **Postdoctoral Associate, Department of Chronic Disease Epidemiology, Yale School of Public Health, Yale University, New Haven, USA**

Supervisor: Shi-Yi Wang (Associate Professor)

- Developed a stochastic state-transition microsimulation model of MGUS-to-multiple myeloma progression
- Estimated transition parameters using SEER-Medicare data and incorporated individual-level risk factors to capture heterogeneous disease progression
- Validated model outputs against SEER incidence data (2014–2019).
- Quantified the population-level impact of screening strategies, highlighting trade-offs between early detection and overdiagnosis.

Current Research Project

Racial/ethnic disparities in lung cancer outcomes: the role of LDCT screening and treatment access

- *Problem:* Racial/ethnic gaps in lung cancer late-stage diagnosis and mortality persist, but it is unclear how much is driven by screening uptake, adherence, or access to novel treatments.
- Developing a race/ethnicity- and histology-specific microsimulation of screening-eligible adults (50–80, ≥20 pack-years), calibrated to SEER incidence and survival.
- Designing factorial scenarios (equal, observed, and counterfactual implementation) to decompose disparities and identify high-impact policy targets. ***In preparation.***

Education

2019 – 2022 **Doctor of Philosophy (Information Science) in the Discipline of Computer and Mathematical Sciences (Mathematical Epidemiology)**
Tohoku University (ranking 3rd in Japan, 23rd in Asia, 79th in world), Sendai, Japan)

2014 – 2016 **Master of Philosophy, Applied Mathematics**
Bacha Khan University, Charsadda, Khyber Pakhtoonkhwa, Pakistan

2011 – 2013 **Master of Science, Mathematics**
Malakand University, Chakdarra, Khyber Pakhtoonkhwa, Pakistan

2009 – 2011 **Bachelor of Science, Mathematics**
Government Degree College Charsadda, Khyber Pakhtoonkhwa, Pakistan

Dissertations

Doctor of Philosophy **Population Dynamics Model for the Effect of Isolation on Final Epidemic Size.**
Supervisor: *Professor Dr. Sci., Hiromi SENO*

Master of Philosophy **A Model for Batch Crystallization of Polymorph with Fine Dissolution Term**
Supervisor: *Assistant Professor Dr. Safyan Mukhtar*

Publications

1 **I. Ahmad**, R. Wang, N. Neparidze, J. Lange, and S.-Y. Wang, "Potential benefits and harms of screening strategies for monoclonal gammopathy of undetermined significance in the us: A simulation study," *Clinical Lymphoma Myeloma and Leukemia*, 2026, ISSN: 2152-2650. DOI: https://doi.org/10.1016/j.clml.2026.06.001.

2. **Ishfaq Ahmad**, C. M. Rutter, C. E. Maerzluft, I. Dengos, K. C. Gogebakan, and J. M. Lange, "Should multi-cancer early detection testing replace guideline-recommended colorectal cancer screening? a comparative modeling analysis," *medRxiv (preprint); submitted to Journal of the National Cancer Institute (JNCI)*, 2026. DOI: https://doi.org/10.64898/2026.07.10.26357782.

3. J. Lange, **Ishfaq Ahmad**, I. J. Dengos, A. Ryan, S. Apostolidou, A. Gentry-Maharaj, S. Holloway, M. D. Ryser, U. Menon, and R. Etzioni, "Can ovarian cancer screening work? a secondary analysis of the uk collaborative trial of ovarian cancer screening," *British Journal of Cancer*, 2026. DOI: https://doi.org/10.1038/s41416-026-03490-2.

4. P. Theprungsirikul, R. Wang, **Ishfaq Ahmad**, N. Neparidze, X. Ma, S.-H. Chang, and S.-Y. Wang, "Association between patient demographics and smoldering multiple myeloma progression to multiple myeloma: A seer-medicare data analysis," *Clinical Lymphoma Myeloma and Leukemia*, 2025, ISSN: 2152-2650. DOI: 10.1016/j.clml.2025.06.016.

5. W. Rahman, **I. Ahmad**, and M. Khan, "A mathematical modeling framework for analyzing and optimizing education systems," *Education Science and Management*, 2024. DOI: https://doi.org/10.56578/esm020405.

6. Y. Xie, **I. Ahmad**, T. Ikpe, E. Sofia, and H. Seno, "What influence could the acceptance of visitors cause on the epidemic dynamics of a reinfectious disease?: A mathematical model," *Acta Biotheoretica*, vol. 72, no. 3, 2024. URL: https://doi.org/10.1007/s10441-024-09478-w.

7. **I. Ahmad** and H. Seno, "An epidemic dynamics model with limited isolation capacity," *Theory in Biosciences*, vol. 142, pp. 259–273, 2023. DOI: 10.1007/s12064-023-00399-9.

8. S. Mukhtar, S. Khan, **I. Ahmad**, and M. Sohaib, "A numerical approach to solve volume-based batch crystallization model with fines dissolution unit," *Processes*, vol. 7, p. 453, 2019. DOI: 10.3390/math8050729.

Manuscripts in Progress

I. Ahmad, J. Lange. Estimating Other-Cause Mortality for Multi-Cancer Early Detection Life Tables, 2026. *In preparation.*

J. Lange, **I. Ahmad.** MCEDsim: a simulation platform for projecting the population impact of multicancer early detection testing, 2026. *In preparation.*

Research Software

MCEDsim: Simulation Platform for Multicancer Early Detection Testing
An **R package** for simulating the population-level benefit and harms of multicancer early detection (MCED) testing. Features pre-fitted natural history models calibrated to US SEER incidence data across 15 solid cancer sites, with a flexible simulation engine to project outcomes across varying test sensitivities, detectable pre-clinical windows, and screening strategies.
Institution: Oregon Health & Science University github.com/Ishfaq8899/MCEDsim

Othercause: Other-Cause Mortality Estimation for MCED Life Tables
An **R package** for generating reproducible, age-, sex-, and year-specific estimates of other-cause mortality for use in MCED life table analyses.
Institution: Oregon Health & Science University github.com/Ishfaq8899/Othercause

Grant History

Nov 2024 - June 2026 — **Cancer Early Detection Advanced Research (CEDAR) Full Project (2024-1960)**
Title: Evaluating of multicancer early detection testing vs traditional colorectal cancer screening: a modeling approach

Sep 2023 – Mar 2025 — **Cancer Early Detection Advanced Research (CEDAR) Full Project (2022-1590)**
Title: Model of ovarian cancer as a research resource (MOOVRR)

Aug 2023 – Aug 2024 — **Cancer Intervention and Surveillance Modeling Network (CISNET) Junior Investigator Career Enhancement (JUICE) Pilot Grant**
Title: Incorporating smoldering myeloma into multiple myeloma natural history modeling: SEER-Medicare data analysis and simulation study
Grant amount: 20,000 USD

Collaborative Research Projects

- **A behavioral epidemic model of hepatitis C transmission with syringe reuse dynamics**
  - *Problem:* Syringe reuse sustains HCV transmission among people who inject drugs (PWID), yet most models ignore how people adopt and quit this behavior.
  - Built an SIR+U model with a behavioral "syringe reuser" compartment and dual (direct and syringe-mediated) transmission; fitted to US acute HCV surveillance data (2015–2021).
  - Found that behavior adoption and cessation rates are key drivers of HCV persistence, supporting harm-reduction strategies. ***Submitted to The European Physical Journal Plus.***
- **Modeling the impact of rehabilitation on hepatitis C transmission among PWID**
  - *Problem:* Rehabilitation may reduce some transmission routes more than others, and relapse can undo its benefits.
  - Extended the SIR+U model with a rehabilitation compartment, separate risk reductions for direct and syringe-based transmission, and relapse to high-risk behavior. ***In preparation.***

Conferences/Workshops

As a Speaker

Oct 21-24, 2024 — **Early Detection of Cancer Conference 2024**
Title: Potential benefits and harms of screening monoclonal gammopathy of undetermined significance in the US
Location: *Hyatt Regency San Francisco Downtown SOMA 50 3rd St, San Francisco, CA, 94103*
https://www.earlydetectionresearch.com/

Dec 12-15, 2023 — **Cancer Intervention and Surveillance Modeling Network (CISNET) 2023 Annual Meeting**
Title: Benefits and harms of screening monoclonal gammopathy of undetermined significance in the US
Location: *NCI Shady Grove Campus, 9609 Medical Center Drive Rockville, MD 20850*
https://www.scgcorp.com/cisnetannual2023/Agendas

Oct 22-25, 2023 — **Society for Medical Decision Making (SMDM) 45th Annual North American Meeting**
Title: Benefits and harms of screening monoclonal gammopathy of undetermined significance in the US.
Location: *Philadelphia, Pennsylvania, USA.*
https://smdm.org/uploads/meeting_files/2023_SMDM_Poster_Schedule.pdf

June 26-30, 2023 **Cancer Intervention and Surveillance Modeling Network (CISNET) 2023 Mid Year Meeting**
Title: Estimating over-diagnosis if screening of monoclonal gammopathy of undetermined significance in the US
Location: *Fred Hutchinson Cancer Center, 1100 Fairview Avenue North Seattle, WA 98109*
https://www.scgcorp.com/cisnetmidyear2023/Agendas

Feb 02, 2022 **Aobayama Colloquium for Mathematical Biology, 2022**
Title: How does the final epidemic size depend on the limited isolation capacity?: Consideration with a mathematical model
Location: *Department of Computer and Mathematical Sciences, Graduate School of Information Sciences, Tohoku University Aramaki-Aza-Aoba 6-3-09, Aoba-ku, Sendai, Miyagi, 980-8579, Japan*
https://www.math.is.tohoku.ac.jp/~seno/seminars/abcMB/abcMB.html

Sep 13-15, 2021 **31st Annual Meeting of Japanese Society for Mathematical Biology (JSMB), 2021**
Title: A Model for the Effect of Limited Isolation Capacity on the Final Epidemic Size
Location: *University of Miyazaki, Kibana campus, Miyazaki, Japan*
https://sites.google.com/view/jsmb2021conference-eng/schedule-program?authuser=0

Jan 27, 2021 **Aobayama Colloquium for Mathematical Biology, 2021.**
Title: The longer isolation may cause the increase of final epidemic size: Consideration with a mathematical model
Location: *Department of Computer and Mathematical Sciences, Graduate School of Information Sciences, Tohoku University Aramaki-Aza-Aoba 6-3-09, Aoba-ku, Sendai, Miyagi, 980-8579, Japan*
https://www.math.is.tohoku.ac.jp/~seno/seminars/abcMB/abcMB.html

As a Attendee

May 21-24, 2024 **Cancer Intervention and Surveillance Modeling Network (CISNET) 2023 Annual Meeting**
Location: *NCI Shady Grove Campus, 9609 Medical Center Drive Rockville, MD 20850*
https://events.cancer.gov/nci/cisnet

Feb 8-11, 2022 **The 13th International Conference on Dynamical Systems Applied to Biology and Natural Sciences (DSABNS) 2022**
Location: *Basque Center of Applied Mathematics, Bilbao, Spain*

Jan 11-13, 2021 **International Conference for Mathematical Modeling and Applications, 2021**
Location: *Meiji University, Japan*
http://cmma.mims.meiji.ac.jp/eng/events/ICMMA2021/index.html

July 22-27, 2021 **The 21th Computational and Mathematical Methods in Science and Engineering (CMMSE 21)**
Location: *Amsterdam, Berlin, Rome, Stockholm, Vienna*

July 5-7, 2021 **Toward an integration of fluids, ecology, and evolution of Biofluids 2021, Research Institute for Mathematical Sciences. (Online)**
Location: *Kyoto University, Japan.*

Mar 9-10, 2021 **Modeling infectious disease: COVID-19 and beyond. (Online)**
Title: An age-structured epidemic model recognizing waning and boosting of immune status.
Title: Effective epidemic model for COVID-19 using accumulated deaths.
Title: When does a second wave appear in an epidemic?
Title: Evaluation of the epidemic prevention effect of non-pharmaceutical interventions for COVID-19 in Japan.

Feb 2-5, 2021 — **The 12th International Conference on Dynamical Systems Applied to Biology and Natural Sciences (DSABNS) 2021. (Online)**
Location: *Basque Center of Applied Mathematics, Spain.*

Dec 22, 2020 — **Mathematical model-based quantitative data analysis for COVID-19, iTHEMS-Biology Seminar. (Online)**
Location: *Kyushu University, Japan.*
Guest speaker: *Professor Shingo Iwami*

Sep 20-22, 2020 — **Annual Meeting of the Japanese Society for Mathematical Biology, JSMB2020 (Online)**
Location: *Nagoya University, Japan.*

October 07-08, 2019 — **AOBA-YAMA COLLOQUIUM FOR MATHEMATICAL BIOLOGY**
Location: *Tohoku University, Sendai, Japan.*
Guest speaker: *Professor Horst Malchow (Osnabruck University Germany)*
Guest speaker: *Professor Takenori Takada (Hokkaido University)*

Awards

Aug 2023 – Aug 2024 — **Junior Investigator Career Enhancement (JUICE) Pilot Grant (20,000 USD)**
Source: *Cancer Intervention and Surveillance Modeling Network (CISNET).*
Incorporating Smoldering Myeloma into Multiple Myeloma Natural History Modeling: SEER-Medicare Data Analysis and Simulation Study

2019–2022 — **Scholarship for PhD Studies**
Source: *Ministry of Education, Culture, Sports, Science and Technology (MEXT) Scholarship.*
Event: *I was among the twelve Pakistan students who won the University-Recommended MEXT Scholarship of Japan for PhD studies in 2019.*

2014–2016 — **Scholarship for Master Studies**
Source: *Prime Minister's Fee Reimbursement Scheme.*

2015 — **Haier Laptop for Research Studies**
Source: *Prime Minister's National Laptop Scheme.*

Courses Studied during Ph.D

April 2021 — **An Introduction to Mathematical Epidemiology, Second Edition.**
Maia Martcheva

Oct 2020 — **Mathematical Models in Population Biology and Epidemiology, Second Edition.**
Fred Brauer, and Carlos Castillo-Chavez

April 2020 — **A First Course in Differential Equations, Third Edition.**
J. David Logan

Oct 2019 — **An Introduction to Mathematical Population Dynamics, along the Trail of Volterra and Lotka.**
Mimmo Iannelli, and Andrea Pugliese

Reviewer

2023 – 2026 — **American Journal of BioScience**
Journal of Mathematics and Computer Science (JMCS)
Journal of Mathematics and Statistics

Mathematical Biosciences

Teaching Experience

2017–2019 **Lecturer of Mathematics**
New Muslim College, Charsadda, Pakistan

2016–2017 **Lecturer of Mathematics**
Government Post Graduate College, Charsadda, Pakistan

- *Calculus I, II & III, Linear Algebra, Differential Equations*
- *Numerical Analysis, Mathematical Modeling, Dynamical Systems*

Languages

English Alternative language and medium of instruction (Reading, Writing, Listening, and Speaking)

Urdu National language (Reading, Writing, Listening, and Speaking)

Japanese Beginner

References

Jane Lange (Postdoc Mentor)
Position: Staff Scientist
University: Oregon Health Science University
Address: Knight Cancer Institute's, Oregon Health Science University, Portland, United States
Email address: langeja@ohsu.edu

Shi-Yi Wang (Former Postdoc Mentor)
Position: Associate Professor
University: Yale University
Address: Department of Chronic Disease Epidemiology, Yale School of Public Health, Yale University, Connecticut, United States.
Email address: shiyi.wang@yale.edu

Hiromi SENO (PhD Mentor)
Position: Professor, Dr.Sci.
University: Tohoku University
Address: Research Center for Pure and Applied Mathematics, Department of Computer and Mathematical Sciences, Graduate School of Information Sciences, Tohoku University, Aramaki-Aza-Aoba 6-3-09, Aoba-ku, Sendai, Miyagi, 980-8579, Japan
Email address: seno@math.is.tohoku.ac.jp